# Factorization into A = LU

Record elimination once as $A = LU$, then solve for any new target with two quick triangular passes. Plus what elimination costs, and $PA = LU$ when rows must be swapped.

## PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.


1. **Inverse of a product**
   $(AB)^{-1} = B^{-1}A^{-1}$: undo in reverse order. Plus the transpose.
2. **2 × 2: from EA = U to A = LU**
   Move the elimination matrix to the other side and $L$ appears.
3. **3 × 3: the multipliers go straight into L**
   The three-fund portfolio: $L$ is just the multipliers 12, 1, 0.2, with no messy mixing.
4. **Pulling out the pivots: A = LDU**
   Split the pivots into a diagonal matrix $D$.
5. **Solving with L and U**
   Two triangular solves. New targets reuse the same $L$ and $U$.
6. **How expensive is elimination?**
   About $n^3/3$ steps to factor, then only $n^2$ per new target.
7. **Row exchanges: PA = LU**
   When a row swap is needed, and the family of permutation matrices.
8. **Singular matrices and LU**
   A singular matrix can still have $A = LU$: the zero pivot sits at the bottom of $U$.


METHODS IN THIS LESSON

Inverse of a product · LU decomposition (Doolittle) · Gauss–Jordan (preview) · LDU decomposition · Forward & back substitution · Operation count · PA = LU

## PART 2 Prerequisites

This lesson finishes elimination, so it leans on Lesson 2: Elimination with Matrices. Click any link to review it, then come back.

- The 3 × 3 portfolio, eliminated step by step: multipliers 12, 1, 0.2 and $U$.
- Elimination matrices: $E_{32}E_{31}E_{21}A = U$, and why order matters.
- Undoing a step: $E_{21}^{-1}$ adds back what $E_{21}$ subtracted.
- Permutation matrices and when elimination breaks down.
- The three-fund portfolio and its "change only the target" example (Lesson 1).

Lesson 3 (Multiplication and Inverse Matrices) covers inverses of whole matrices. Here we only need inverses of single elimination steps, which Lesson 2 already gave us.

## PART 3 Explanation

**THE RUNNING EXAMPLE: THE PORTFOLIO, FACTORED ONCE**

In Lesson 2 we eliminated the three-fund portfolio and reached $U$:

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix}}_{A} \xrightarrow{\ell_{21}=12,\ \ell_{31}=1,\ \ell_{32}=0.2} \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 0 & 0 & -0.4 \end{bmatrix}}_{U}$$

Now think like a fund desk: the **funds and rules stay the same** ($A$ is fixed) but **clients bring different targets** every day: ₹10 lakh at 11.1%, ₹10 lakh at 9.8%, ₹15 lakh... Redoing elimination for each is wasteful. This lesson records elimination once, as $A = LU$, and reuses it.

## 1. Inverse of a product

**RULE · INVERSE AND TRANSPOSE OF A PRODUCT**

$(AB)^{-1} = B^{-1}A^{-1}$ and $(AB)^T = B^T A^T$: reverse the order. Also $(A^{-1})^T = (A^T)^{-1}$.

Strang starts with one basic fact, because $L$ will turn out to be a product of inverses.

INVERSE OF A PRODUCT

If $A$ and $B$ are invertible, then $(AB)^{-1} = B^{-1}A^{-1}$: the same inverses, **in reverse order**.

**Check it:** $(AB)(B^{-1}A^{-1}) = A\,(BB^{-1})\,A^{-1} = A\,I\,A^{-1} = AA^{-1} = I$. We only moved parentheses (Lesson 2), and the middle pair cancelled first.

SHOES AND SOCKS

You put on socks, then shoes. To undo it you take off the shoes first, then the socks. Undoing always goes in reverse order. In elimination: we applied $E_{21}$, then $E_{31}$, then $E_{32}$; to undo them we apply $E_{32}^{-1}$ first and $E_{21}^{-1}$ last.

The transpose $A^T$ swaps rows and columns: the entry in row $i$, column $j$ of $A$ is in row $j$, column $i$ of $A^T$. For our two-fund matrix:

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} \qquad A^T = \begin{bmatrix} 1 & 12 \\ 1 & 7 \end{bmatrix}$$

In $A$ each **column** is a fund; in $A^T$ each **row** is a fund. Transposes follow the same reverse-order rule, $(AB)^T = B^T A^T$, and for an invertible matrix $(A^{-1})^T = (A^T)^{-1}$: you may transpose and invert in either order. Transposes return in Lesson 5.

▼ Check yourself: what is $(ABC)^{-1}$?

$C^{-1}B^{-1}A^{-1}$. Check: $ABC\,C^{-1}B^{-1}A^{-1}$ collapses from the middle outwards to $I$.

## 2. 2 × 2: from EA = U to A = LU

METHOD · LU DECOMPOSITION (LU FACTORIZATION)

Write $A = LU$: $L$ lower triangular with 1s on its diagonal and the elimination multipliers below; $U$ upper triangular with the pivots. It is Gaussian elimination, recorded. This form, with the 1s in $L$, is also called the **Doolittle** factorization (the **Crout** form puts the 1s in $U$ instead).

The two-fund portfolio took one elimination step ($\ell_{21} = 12$):

$$\underbrace{\begin{bmatrix} 1 & 0 \\ -12 & 1 \end{bmatrix}}_{E_{21}} \underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & -5 \end{bmatrix}}_{U}.$$

That has the elimination matrix on the left of $A$. Strang wants $A$ alone on one side. Multiply both sides by $E_{21}^{-1}$, which **adds back** 12 × row 1 (Lesson 2):

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}}_{L=E_{21}^{-1}} \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & -5 \end{bmatrix}}_{U}.$$

**Check:** row 2 of $LU$ is $12 \times (1,1) + 1 \times (0,-5) = (12,7)$ ✓, the return rule.

> **A = LU**
> $U$ is **upper** triangular with the **pivots** on its diagonal. $L$ is lower triangular with **1s** on its diagonal and the **multipliers** below it. "L" for lower, "U" for upper.

In words: *the return rule = 12 × (budget rule) + (the eliminated rule $-5y = -15$).* $L$ records how to rebuild each original rule from the rows of $U$.

> ▼ **Check yourself (Strang's example): factor $A = \left[\begin{smallmatrix} 2 & 1 \\ 8 & 7 \end{smallmatrix}\right]$.**
>
> $\ell_{21} = 8/2 = 4$, so $U = \left[\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right]$ and $L = \left[\begin{smallmatrix} 1 & 0 \\ 4 & 1 \end{smallmatrix}\right]$. Check: $4(2,1) + (0,3) = (8,7)$ ✓.

## 3. 3 × 3: the multipliers go straight into L

For the three-fund portfolio, Lesson 2 gave $E_{32}E_{31}E_{21}A = U$. Undo the three steps **in reverse order** (topic 1):

$$A = E_{21}^{-1}\,E_{31}^{-1}\,E_{32}^{-1}\,U = LU.$$

Each inverse just flips the sign of its multiplier, and the product comes out remarkably clean:

$$\underbrace{\begin{bmatrix}1&0&0\\12&1&0\\0&0&1\end{bmatrix}}_{E_{21}^{-1}}\underbrace{\begin{bmatrix}1&0&0\\0&1&0\\1&0&1\end{bmatrix}}_{E_{31}^{-1}}\underbrace{\begin{bmatrix}1&0&0\\0&1&0\\0&0.2&1\end{bmatrix}}_{E_{32}^{-1}}=\underbrace{\begin{bmatrix}1&0&0\\\mathbf{12}&1&0\\\mathbf{1}&\mathbf{0.2}&1\end{bmatrix}}_{L}$$

$$\underbrace{\begin{bmatrix}1&1&1\\12&7&15\\1&0&1.2\end{bmatrix}}_{A}=\underbrace{\begin{bmatrix}1&0&0\\12&1&0\\1&0.2&1\end{bmatrix}}_{L}\underbrace{\begin{bmatrix}1&1&1\\0&-5&3\\0&0&-0.4\end{bmatrix}}_{U}$$

**Check row 3:** $1\times(1,1,1)+0.2\times(0,-5,3)+1\times(0,0,-0.4)=(1,1-1,1+0.6-0.4)=(1,0,1.2)$ ✓, the exposure rule.

> **THE KEY FACT OF THIS LESSON**
>
> With no row exchanges, **the multipliers go directly into $L$:** $\ell_{21}=12,\ell_{31}=1,\ell_{32}=0.2$ sit in exactly their positions. To get $L$ there is no work at all, just keep a record of the multipliers as you eliminate.

### Why L is clean but E is messy

Compare with the single elimination matrix from Lesson 2:

$$E=E_{32}E_{31}E_{21}=\begin{bmatrix}1&0&0\\-12&1&0\\\mathbf{1.4}&-0.2&1\end{bmatrix}\qquad\text{vs}\qquad L=E^{-1}=\begin{bmatrix}1&0&0\\12&1&0\\\mathbf{1}&0.2&1\end{bmatrix}$$

Where does $E$'s $1.4$ come from? Going **forward**, row 3 had 1 × row 1 removed directly, and then lost 0.2 × the *new* row 2, which itself was "row 2 − 12 × row 1". That second step quietly **added back** $0.2\times12=2.4$ of row 1. Net effect on row 1: $-1+2.4=+1.4$. The steps interfere.

Going **backward** with $L$, nothing interferes: each original row is simply "its multipliers × the rows of $U$ above it, plus its own row of $U$". Strang: "the multipliers just sit in the matrix $L$." This is why software stores $L$ and $U$, never $E$.

> ▼ **Check yourself (MIT problem 4.1): find $L$ and $U$ for $A=\begin{bmatrix}1&3&0\\2&4&0\\2&0&1\end{bmatrix}$.**
>
> Multipliers: $\ell_{21}=2$ gives row 2 $=(0,-2,0)$; $\ell_{31}=2$ gives row 3 $=(0,-6,1)$; $\ell_{32}=-6/-2=3$ gives row 3 $=(0,0,1)$. So $U=\begin{bmatrix}1&3&0\\0&-2&0\\0&0&1\end{bmatrix}$ and $L=\begin{bmatrix}1&0&0\\2&1&0\\2&3&1\end{bmatrix}$, the multipliers 2, 2, 3 in place.

METHOD · GAUSS–JORDAN ELIMINATION (PREVIEW)

Eliminate **above** each pivot as well as below, turning $[A \mid I]$ into $[I \mid A^{-1}]$. MIT's solution to problem 4.1 finds $E^{-1} = L$ this way. We don't need it here, because $L$ can be read straight off the multipliers. Full treatment in Lesson 3.

## 4. Pulling out the pivots: A = LDU

METHOD · LDU DECOMPOSITION

$A = LDU$: unit lower triangular $L$, diagonal $D$ holding the pivots, unit upper triangular $U$. The same information as $LU$, arranged symmetrically.

$L$ has 1s on its diagonal but $U$ has the pivots. To balance them, divide each row of $U$ by its pivot and collect the pivots in a diagonal matrix $D$:

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 0 & 0 & -0.4 \end{bmatrix}}_{U} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -0.4 \end{bmatrix}}_{D} \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -0.6 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{new } U}$$

Row 2 of $U$ divided by $-5$ is $(0, 1, -0.6)$; row 3 divided by $-0.4$ is $(0, 0, 1)$. Now $A = LDU$, with 1s on the diagonals of both triangular factors and the pivots $1, -5, -0.4$ in the middle. Strang mostly stays with $A = LU$; $LDU$ is the same information arranged symmetrically.

## 5. Solving with L and U

METHOD · FORWARD AND BACK SUBSTITUTION

**Forward substitution** solves the lower triangular $Lc = b$ from the **top** row down. **Back substitution** then solves $Ux = c$ from the **bottom** row up. Together they solve $Ax = b$ in about $n^2$ steps once $A = LU$ is known.

Once $A = LU$ is known, solving $Ax = b$ means solving $L(Ux) = b$. Call $Ux = c$ and do it in two easy triangular passes:

1. **Forward:** solve $Lc = b$ from the **top** down. (This is exactly what elimination did to $b$.)
2. **Back:** solve $Ux = c$ from the **bottom** up. (Back substitution, as in Lesson 2.)

### Client 1: ₹10 lakh, ₹1,11,000 return, 7.4 exposure

$b = (10, 111, 7.4)$. **Forward**, $Lc = b$:

- $c_1 = 10$.
- $12c_1 + c_2 = 111$, so $c_2 = 111 - 120 = -9$.
- $1c_1 + 0.2c_2 + c_3 = 7.4$, so $c_3 = 7.4 - 10 + 1.8 = -0.8$.

**Back**, $Ux = c$: $-0.4z = -0.8$ gives $z = 2$; $-5y + 3(2) = -9$ gives $y = 3$; $x + 3 + 2 = 10$ gives $x = 5$. ₹5 lakh Nifty, ₹3 lakh bonds, ₹2 lakh midcap, as before. Notice $c = (10, -9, -0.8)$ is exactly the right-hand side elimination produced in Lesson 2.

### Client 2: same funds, a new target

$b = (10, 98, 5.2)$: ₹10 lakh, ₹98 thousand return, 5.2 exposure. **No new elimination**, just the two passes with the same $L$ and $U$:

- **Forward:** $c_1 = 10$; $c_2 = 98 - 12(10) = -22$; $c_3 = 5.2 - 10 - 0.2(-22) = 5.2 - 10 + 4.4 = -0.4$.
- **Back:** $-0.4z = -0.4$ gives $z = 1$; $-5y + 3(1) = -22$ gives $y = 5$; $x + 5 + 1 = 10$ gives $x = 4$.

₹4 lakh Nifty, ₹5 lakh bonds, ₹1 lakh midcap, the answer Lesson 1 found for this target. Check: $48 + 35 + 15 = 98$ ✓.

### Client 3: ₹15 lakh

$b = (15, 159, 9.6)$. Forward: $c = (15,\ 159 - 180,\ 9.6 - 15 + 4.2) = (15, -21, -1.2)$. Back: $z = 3$, then $-5y + 9 = -21$ so $y = 6$, then $x = 15 - 6 - 3 = 6$. ₹6 lakh, ₹6 lakh, ₹3 lakh.

> **FACTOR ONCE, SOLVE MANY**
> The hard part, elimination on $A$, was done **once**. Each new client costs only two triangular passes. With many targets and the same funds, $A = LU$ is how it's done in practice.

## 6. How expensive is elimination?

> **RULE · OPERATION COUNT: $O(N^3)$ TO FACTOR, $O(N^2)$ TO SOLVE**
> Gaussian elimination / LU on an $n \times n$ matrix takes about $\frac{1}{3}n^3$ multiply-subtract steps; each right-hand side then costs about $n^2$. Read $O(n^3)$ as "grows like $n^3$".

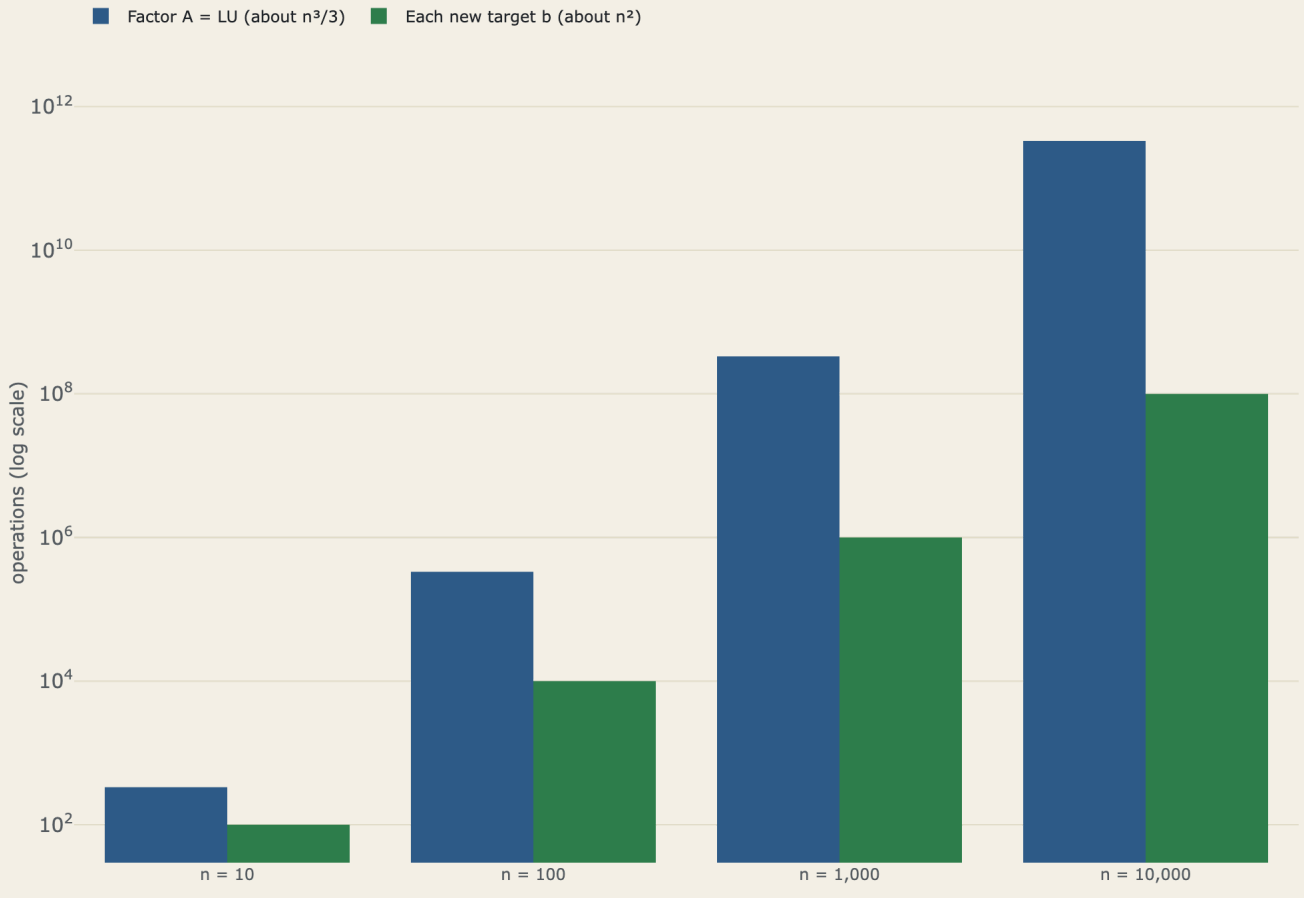
A practical question Strang asks: for an $n \times n$ matrix, how many operations does elimination take? Count one "multiply a row entry and subtract" as one operation.

- Clearing column 1 changes almost every entry below row 1: about $n^2$ operations.
- Column 2 works on a matrix one size smaller: about $(n-1)^2$. Then $(n-2)^2$, and so on down to $1^2$.

$$n^2 + (n-1)^2 + \cdots + 2^2 + 1^2 \;\approx\; \int_0^n x^2\,dx \;=\; \tfrac{1}{3}n^3.$$

THE COST

**Factoring $A = LU$: about $\frac{1}{3}n^3$ operations.** Each right-hand side $b$ (forward + back): about $n^2$. Doubling the size of the problem makes factoring 8 times as expensive, but each new target only 4 times.




**Operations needed (log scale).** For $n = 1000$ (say, 1,000 assets): factoring takes about 333 million operations; each new target only 1 million.

Strang's aside: computing with determinants instead would cost about $n!$ operations, astronomically more. Elimination is the method that scales.

## 7. Row exchanges: PA = LU

> **METHOD · PA = LU (LU WITH ROW EXCHANGES)**
>
> Put the rows in a workable order first with a permutation $P$, then factor: $PA = LU$. Every invertible matrix has one. Software picks $P$ by **partial pivoting** (largest entry in each pivot position) for accuracy.

So far, no row exchanges. What if a 0 lands in a pivot position? Lesson 2's **credit fund** (12% return, beta 0) did exactly that. The fix: do the needed row exchange **first**, then factor. For the credit-fund matrix, swap rows 2 and 3 with $P_{23}$:

$$\underbrace{\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}}_{P_{23}}\underbrace{\begin{bmatrix}1&1&1\\12&12&15\\1&0&1.2\end{bmatrix}}_{A}=\underbrace{\begin{bmatrix}1&0&0\\1&1&0\\12&0&1\end{bmatrix}}_{L}\underbrace{\begin{bmatrix}1&1&1\\0&-1&0.2\\0&0&3\end{bmatrix}}_{U}$$

After the swap the rows are budget, exposure, return. The multipliers are $\ell_{21} = 1$ (exposure − budget), $\ell_{31} = 12$ (return − 12 × budget), and $\ell_{32} = 0$ (that row was already done). They sit in $L$ as before.

> **PA = LU**
>
> Every invertible matrix can be factored as $PA = LU$, where $P$ puts the rows in an order that needs no exchanges during elimination.

### The family of permutation matrices

For 3 × 3 there are exactly **6** permutation matrices, all the ways to reorder the rows of $I$:

$$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$$

The identity (no change), three single swaps, and two cycles that move all three rows. For $n \times n$ there are $n!$ of them: 24 for 4 × 4.

- **Closed family:** multiply two permutations and you get another one (reordering twice is still a reordering). Inverses stay in the family too. Mathematicians call this a *group*.
- **Inverse = transpose:** $P^{-1} = P^T$. A single swap is its own inverse; the two cycles are each other's inverses.

▼ **Check yourself: why is $P^{-1} = P^T$ for $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$?**

$P$ sends row 2 to the top, row 3 to the middle, row 1 to the bottom. $P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ sends them back: $P^T P = I$.

## 8. Singular matrices and LU

A point from the recitation: **a singular matrix can still have an LU factorization.** What fails is not the factoring but the solving. Take Lesson 2's **balanced fund** (half Nifty, half bonds):

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ 1 & 0.2 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & -2.5 \\ 0 & 0 & \mathbf{0} \end{bmatrix}}_{U}$$

Elimination never needed a row exchange, so $L$ and $U$ exist, and $L$ is **identical** to the good portfolio's (the first two columns are the same funds). The difference shows up at the bottom of $U$: a **0 where the third pivot should be**. Forward substitution works ($c = (10, -9, -0.8)$), but back substitution hits $0 \cdot z = -0.8$: no solution.

**WATCH OUT**

"Has an LU factorization" does not mean "invertible". Look at the diagonal of $U$: all pivots non-zero ⇔ invertible.